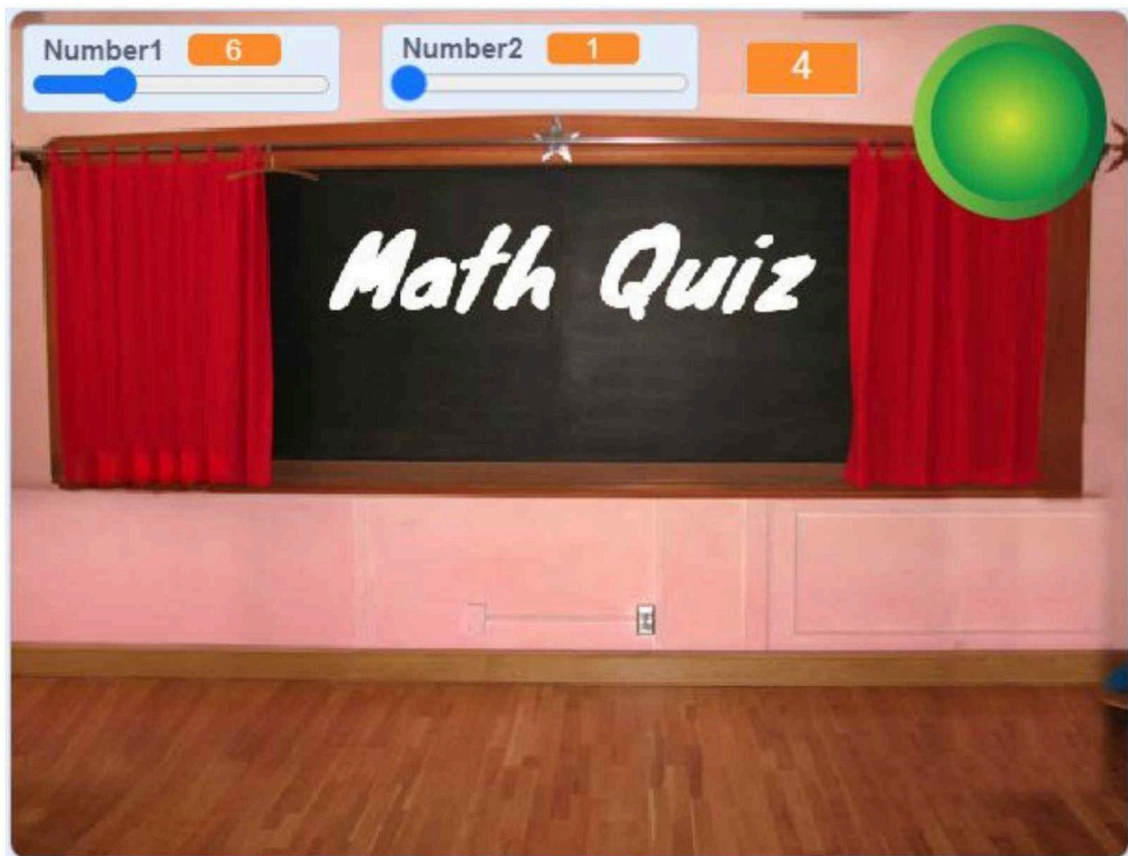


# Unit 3

## Control & Operators Workshop

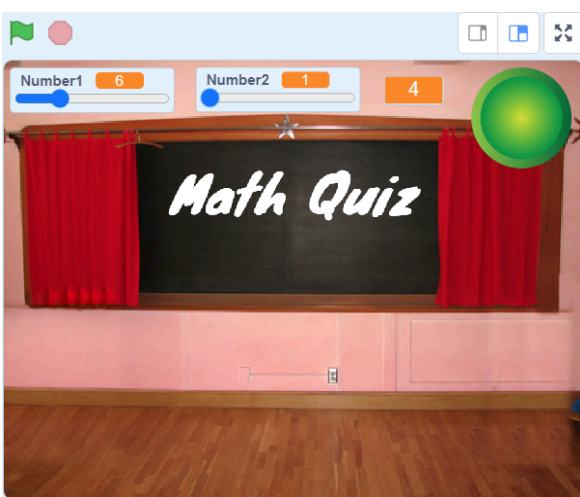


# Math Quiz

## Unit 3: Control and Operator Workshop

Save the file with a new name. Your new name should identify the project.

\* Control and Operators    \* Math Quiz    \* Project 3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Workshop summary</b></p> <p>The Control menu has loops and <i>if/then</i> blocks. The Operations menu supplies most of the diamond blocks used with them. Together, Control and Operations provide the framework for most SCRATCH programs.</p> <p>In this project, we use Control and Operator blocks to build a math game. The game chooses a random value for total. Then, the player is tasked with adjusting two sliders so that their sum equals the total.</p> |                                                                                                                                                                        | <p><b><u>New Terms</u></b></p> <p><b><u>Interface Terms</u></b><br/>condition, slider, click</p> <p><b><u>Code Block Terms</u></b><br/>forever loop, control and operator blocks, repeat loop, if/then, nested if, true/false</p> |
| <p><b>Class Studio:</b><br/><a href="https://tinyurl.com/StchStudio01">https://tinyurl.com/StchStudio01</a></p> <p><b>Video of Full Project:</b> <a href="https://vimeo.com/689105907">https://vimeo.com/689105907</a></p> <p><b>This Worksheet:</b> <a href="https://tinyurl.com/ControlWSPDF">https://tinyurl.com/ControlWSPDF</a></p>                                                                                                                                    | <p><b>Starter Project:</b><br/><a href="https://tinyurl.com/StchStarter03">https://tinyurl.com/StchStarter03</a></p> <p><b>Full Project:</b><br/><a href="https://scratch.mit.edu/projects/660514075">https://scratch.mit.edu/projects/660514075</a></p> |                                                                                                                                                                                                                                   |

### Mini-Challenges in this Lesson

Click on the mini-challenge link to go directly to that section.

[Mini-Challenge 1: Set up the Math problem](#)

[Mini-Challenge 2: Check the Player's Math](#)

[Mini-Challenge 3: Give the Player Feedback on the Answer](#)

[Mini-Challenge 4: Change the Game to Offer Ten Math Problems](#)

[Mini-Challenge 5: Increase the Pressure with a Timer](#)

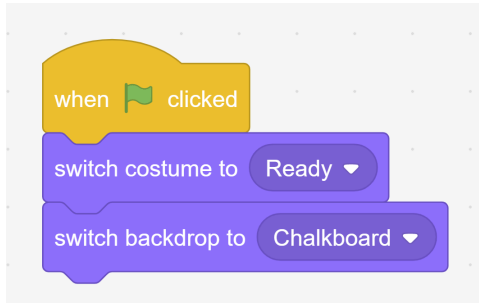
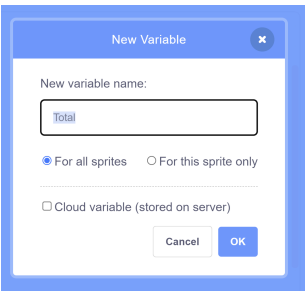
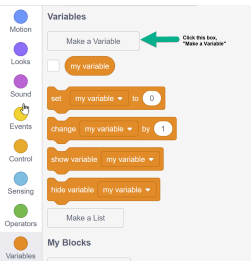
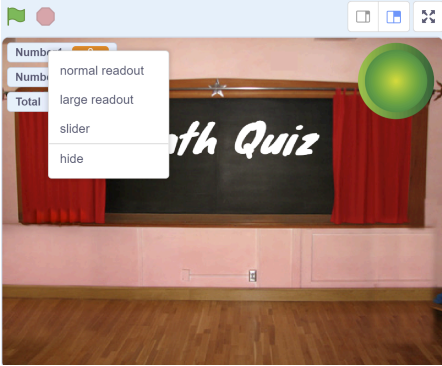
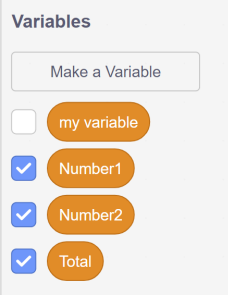
[Mini-Challenge 6: Use a Button to Tell the Result](#)

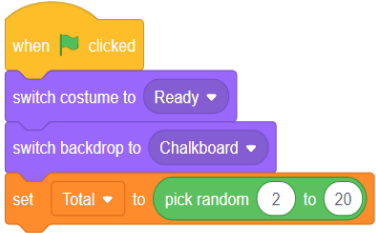
**Mini-Challenge 1: Set Up the Math Problem**

Set up variables to use for the math problem. **Number1** and **Number2** will be added to get **Total**.

**Procedure**

1. Start the project with the *when flag clicked* block.
2. Set the starting values for backdrop and costume as shown in row 1 below.
3. Create three variables: **Number1**, **Number2**, and **Total**.
4. Set the first two as sliders and the third one as a large display.
5. Set the value of Total to be a random number between 2 and 20.

| Menu                     | Code                                                                                                                 | Code block                                                                           | Notes                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Events<br>Looks<br>Looks | when clicked<br><br>switch costume to (Ready)<br><br>set backdrop to (Chalkboard)                                    |    | Put code stack on the <b>BUTTON</b> sprite.                                                                              |
| Variables                | Make a Variable (see note)<br><br>3 Variables:<br>Number1,<br>Number2, and<br>Total                                  |  |                                     |
| Variables                | <b>Variable Settings:</b><br>Right-click to select<br><br>Number1: Slider<br>Number2: Slider<br>Total: Large readout |  | <br>Make variables appear on stage. |

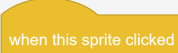
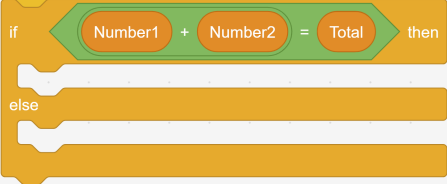
|           |                                         |                                                                                    |                                                                             |
|-----------|-----------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Events    | when flag clicked                       |  | Put code stack on the <b>BUTTON</b> sprite.<br><br>New code is <b>RED</b> . |
| Looks     | switch costume to ready                 |                                                                                    |                                                                             |
| Looks     | switch backdrop to Chalkboard           |                                                                                    |                                                                             |
| Variables | set [ <b>Total</b> ] to ( <u>    </u> ) |                                                                                    |                                                                             |
| Operators | (pick random (2) to (20))               |                                                                                    |                                                                             |

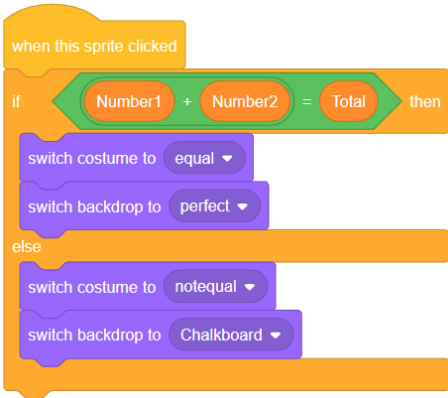
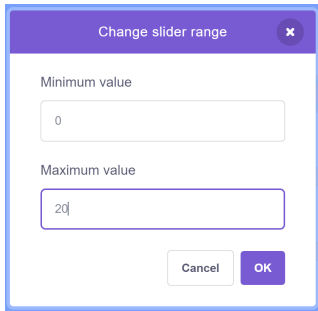
### Mini-Challenge 2: Check the Player's Math

The player can adjust the two slider variables. When the player clicks the button, the game adds the values of the sliders (Number 1 & Number 2). If the result matches the value in Total, the player is notified that the sliders are set correctly.

#### Procedure

1. Add a *when this sprite is clicked* block.
2. Add an *if/then...else* block.
3. Add an *equals* (      =      ) operator.
  - a. On the right side, drop the **total** variable.
  - b. On the left side, drop an **addition** operator.
  - c. Drop the **Number1** and **Number2** variables in the addition operator fields.
4. Use the sprite costume and stage backdrop to show whether the answer is right or wrong.
  - a. The "if" arm gets a costume switch and backdrop switch for a correct answer.
  - b. The "else" arm gets a costume switch and backdrop switch for an incorrect answer.

| Menu          | Code                                         | Code block                                                                           | Notes                               |
|---------------|----------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------|
| Events        | when this sprite is clicked                  |   | This is a second code stack.        |
| Control       | if / then... else                            |  | It is also on <b>BUTTON</b> sprite. |
| Operators     | equals block ( <u>    </u> = <u>    </u> )   |                                                                                      |                                     |
| Variables     | addition block ( <u>    </u> + <u>    </u> ) |                                                                                      |                                     |
| Operators     | Number1 + Number2                            |                                                                                      |                                     |
| Variables (2) |                                              |                                                                                      |                                     |

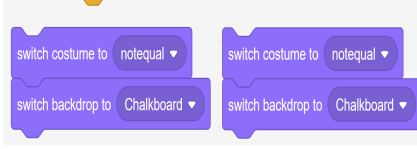
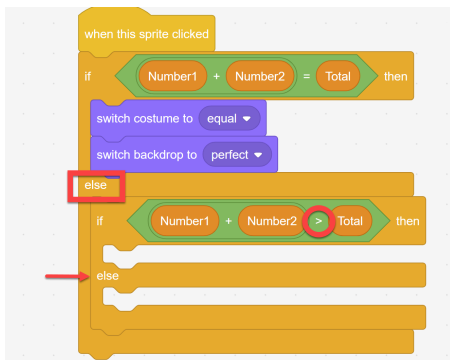
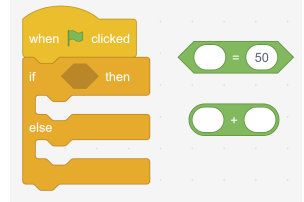
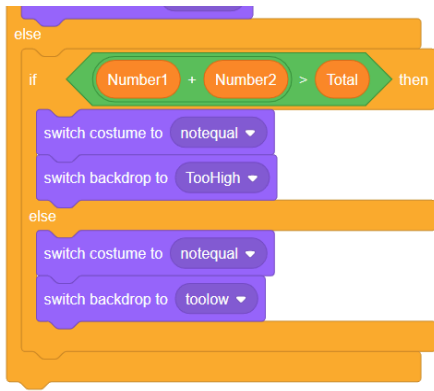
|                                                                                                    |                                                                                                                                                                                                                                                    |                                                                                     |                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Events<br>Control<br>Operators<br>Looks, set 1<br>Looks, set 1<br><br>Looks, set 2<br>Looks, set 2 | when this sprite is clicked<br>if / then . . .<br>equals block ( <u>  </u> + <u>  </u> = <u>Total</u> )<br>switch costume to (equal)<br>switch backdrop to (perfect)<br>else...<br>switch costume to (notEqual)<br>switch backdrop to (Chalkboard) |   | New code is <b>RED</b> .<br>This block stack goes inside the “if / then” statement.<br><b>CORRECT ANSWER</b><br>The second Looks stack belongs in the “else” statement.<br><b>INCORRECT ANSWER</b> |
|                                                                                                    | Change your sliders so they have a range that matches your total.                                                                                                                                                                                  |  |                                                                                                                                                                                                    |

### Mini-Challenge 3: Give the Player Feedback on the Answer

This game can be improved by providing feedback to the player about a wrong answer. We do this by adding another *if / then... else* block in the *else* arm of the code stack from Mini-Challenge 3.

#### Procedure

1. Remove the two blocks from inside the *else* arm of the *when sprite is clicked* code stack and duplicate.
2. Add a new *if/then...else* block in the *else* arm.
3. Add a *greater-than* operator to the diamond block of the *if / then... else*.
  - a. On the right side, drop the **total** variable.
  - b. On the left side, drop an addition operator.
  - c. Drop **Number1** and **Number2** variables into the addition operator fields.
4. Use the sprite costume and stage backdrop to show whether the answer is high or low.
  - a. The *if* arm gets switch costume and switch backdrop for a high answer.
  - b. The *else* arm gets switch costume and switch backdrop for a low answer.

| Menu                                                                                      | Code                                                                                                                                                                                                                                                  | Code block                                                                                             | Notes                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Looks<br>Looks                                                                            | <b>REMOVE</b> from the <i>else</i> arm<br>switch costume to (notEqual)<br>switch costume to (Chalkboard)                                                                                                                                              | <br><b>DUPLICATE</b> | Put code stack on the <b>BUTTON</b> sprite.                                                                                                                                                          |
| Events<br>Control<br>Operators<br>Looks<br>Looks<br>Control<br>Operators<br>Variables (2) | when this sprite is clicked<br>if < ____ > then<br>< Number1 + Number2 = Total<br>><br>switch costume to (equal)<br>switch costume to (perfect)<br>else...<br>if < greater than (Total) > / then<br>addition block: ( __ + __ )<br>Number1<br>Number2 |                      | New code is <b>RED</b> .<br><br>Place this stack <b>INSIDE</b> the <i>else</i> arm.<br>                           |
| Control<br>Looks, 1<br>Looks, 1<br>Looks, 2<br>Looks, 2                                   | if then ...<br><br>switch costume to (notEqual)<br>switch backdrop to (TooHigh)<br><br>... else<br>switch costume to (notEqual)<br>switch backdrop to (TooLow)                                                                                        |                    | Put code stack on the <b>BUTTON</b> sprite.<br><br>New code is <b>RED</b> .<br><br>Change the description of the backdrops and put them in the two arms of the second <i>if / then ... else</i> arm. |

**Mini-Challenge 4: Change the Game to Offer Ten Math Problems**

Modify the *when flag clicked* code.

1. Add a new variable for **Score**.
2. Set **Score** to 0.
3. Add a repeat (\_\_) times loop.
4. Move the logic to calculate the total into the *repeat until* block.

- Set the [total] to a random number.
- Reset the button costume each time the total is calculated.
- Reset the backdrop each time the total is calculated.

5. Add a *wait* to the repeat (\_\_) times.

Modify the *when the sprite is clicked* code stack.

- If there is a correct answer, change [Score] by 1.

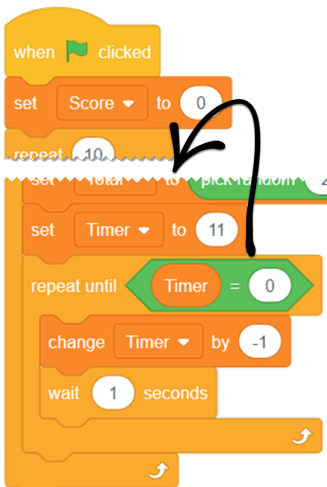
| Menu                                                                                              | Code                                                                                                                                                                                                                                               | Code block | Notes                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variables<br>Events<br>Variables<br>Control<br>Looks<br>Looks<br>Variable/<br>Operator<br>Control | <b>New Variable:</b> Score<br>when flag clicked<br>set [score] to (0)<br>repeat (__)<br><b>MOVE from above</b><br>switch costume to (Ready)<br>switch backdrop to (Chalkboard)<br>set [total] to<br>( pick random 2 to 20 )<br>wait ( __ ) seconds |            | Put code stack on the <b>BUTTON</b> sprite.<br>New code is <b>RED</b> .<br>Alter block stack 1<br><i>when flag clicked</i><br><br>This is the original code stack.                                                                                     |
| Events<br>Control<br>Operators<br>Variable<br>Looks<br>Looks<br>Variable                          | when this sprite clicked<br>if / then. . .<br>Number1 + Number2 = Total<br>switch costume to equal<br>switch backdrop to perfect<br>change [Score] by (__)                                                                                         |            | Put code stack on the <b>BUTTON</b> sprite.<br><b>Alter Block Stack 2</b><br>“When this sprite clicked”<br>New code is <b>RED</b> .<br>Insert a new code block<br>inside the <i>if / then</i> section<br>of the <b>FIRST if / then ... else</b> block. |

**Mini-Challenge 5: Increase the Pressure with a Timer**

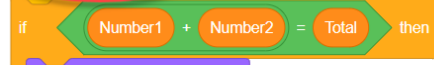
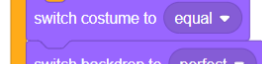
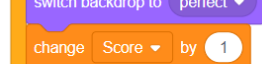
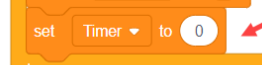
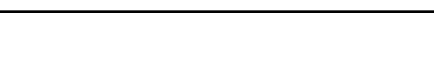
The *repeat (10)* loop controls the game's progress. The programmer knows that the **Total** variable is changed every 10 seconds. The player doesn't know this. A visible timer can be displayed by adding a **Timer** variable and a *repeat until* loop inside the main loop.

**Procedure**

1. Create a new variable **Timer**, and display it on the stage.
2. Add a *set timer to (11)* block.
3. Remove the *wait 10 seconds* block added in the last mini-challenge.
4. Add a *repeat until* loop under the *set timer* block. It will be inside the *repeat (10)* loop.  
This is called a **nested loop!** It replaces the *wait 10 seconds* block.
5. Inside the *repeat until* loop, add a variable and a control block to count down.
  - a. Change timer by -1.
  - b. Wait 1 second.
6. Control the *repeat until* loop by adding an equals operator in the diamond.
  - a. Drop 0 in the right-hand field.
  - b. Drop the timer variable in the left-hand field.

| Menu      | Code                                                       | Code block                                                                           | Notes                                           |
|-----------|------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------|
| Events    | <b>New Variable:</b> Timer<br>(Add ✓ to display on stage.) |  | Put code stack on the <b>BUTTON</b> sprite.     |
| Variable  | when flag is clicked                                       |                                                                                      | New code is <b>RED</b> .                        |
| Variables | set <b>Score</b> to (0)                                    |                                                                                      |                                                 |
| Control   | <b>set [Timer] to 11</b>                                   |                                                                                      |                                                 |
| Operators | <b>repeat until &lt; (Timer) = (0) &gt;</b>                |                                                                                      | <b>Alter Block Stack 1</b><br>when flag clicked |
| Variables | <b>change [Timer] by (-1)</b>                              |                                                                                      |                                                 |
| Control   | <b>wait (1) seconds</b>                                    |                                                                                      |                                                 |

7. Add a *set [Timer] to 0* block.
  - a. This goes after the block where you increased the score.
  - b. Once **Timer** is set to 0, the *repeat until* loop in the other code stack will finish, and the player will get a fresh total to work on.

| Menu      | Code                          | Code block                                                                         | Notes                                       |
|-----------|-------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
| Events    | when this sprite clicked      |   | Put code stack on the <b>BUTTON</b> sprite. |
| Control   | if / then...else              |  |                                             |
| Operators | < Number1 + Number2 = Total > |   | <b>Alter Block Stack 2</b>                  |
| Looks     | switch costume to (equal)     |   | "When this sprite clicked"                  |
| Looks     | switch backdrop to (perfect)  |   |                                             |
| Variable  | change [Score] by (1)         |   |                                             |
| Variable  | set [Timer] to (0)            |  |                                             |

### Mini-Challenge 6: Use the Button to Tell the Result

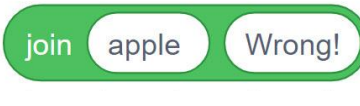
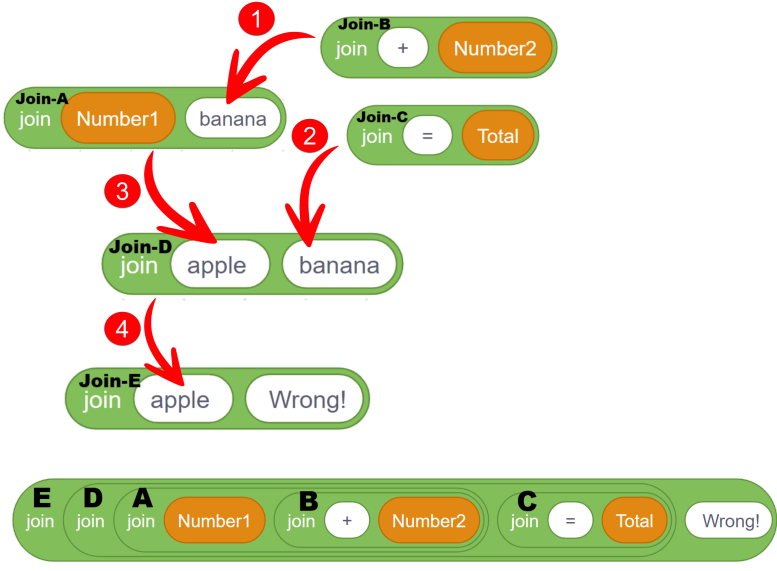
We can use the say (\_\_\_\_) for (\_\_\_\_) seconds in the Looks category to give additional feedback to the player. We could just say something like "Awesome" or "Try Again." But we can use the Join operator to combine words and numbers that we will use in this challenge.

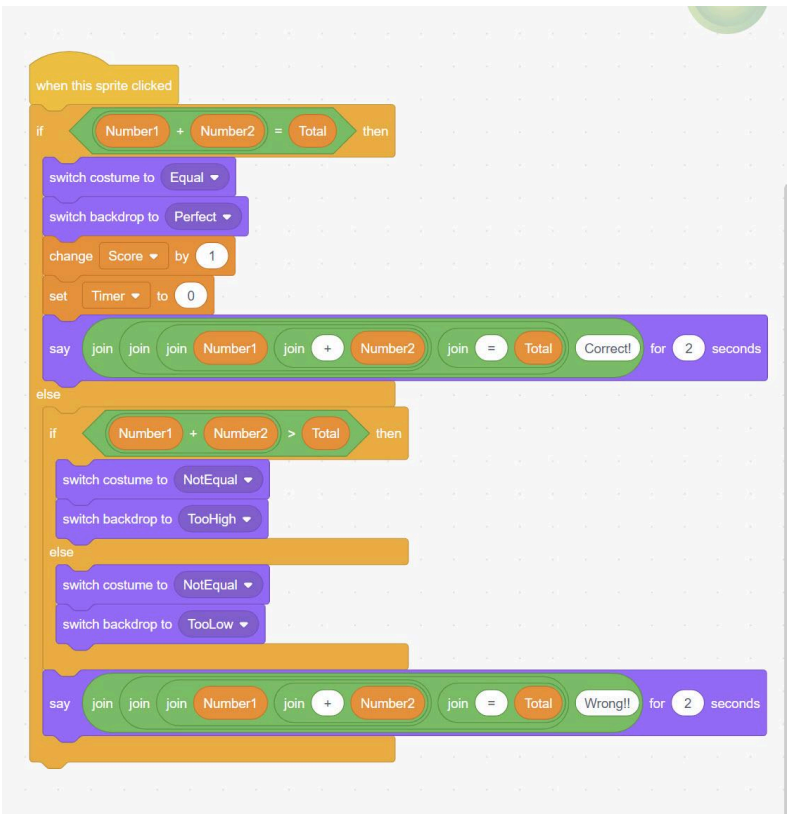
#### **Procedure for Number1 + Number2 = Total Correct! (or Incorrect!)**

1. Pull the say (\_\_\_\_) for (\_\_\_\_) seconds from Looks. Join the string with ALL go inside the first oval.
2. Pull five join operator blocks to use as follows:
  - a. Number1 variable in oval one,
  - b. "+" and Number2 variable,
  - c. (a and b) in the first oval,
  - d. "=" and Total variable.
  - e. Place the entire string of join operators in oval one and (Correct!)
3. Duplicate this join string and change the final join to (Wrong!)

#### **Other possibilities to try**

- High: Number1 + Number2 is too high. Wrong!
- Low: Number1 + Number2 is too low. Wrong!
- Create a variable for the result phrase.
- Use join operators to build the phrase.

| Menu                                                                                                                                                                                                                                                                                                                                                   | Code                                                                                       | Code block                                                                           | Notes                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Operator<br>Variables                                                                                                                                                                                                                                                                                                                                  | Pull <b>5</b> Join (___) (___) blocks.<br>Pull 3 Variables:<br>Number1, Number2, and Total |    | Note: Ignore the words <b>apple</b> and <b>banana</b> . They are just placeholders in the join operator. |
| Operator<br>Variable                                                                                                                                                                                                                                                                                                                                   | Join-A<br>(Number1) and (leave blank)                                                      |    |                                                                                                          |
| Operator<br>Variable                                                                                                                                                                                                                                                                                                                                   | Join-B<br>(space + space) and (Number2)                                                    |    | Space means press the spacebar once.                                                                     |
| Operator<br>Variable                                                                                                                                                                                                                                                                                                                                   | Join-C<br>(space = space) and (Total)                                                      |    | Space means press the spacebar once.                                                                     |
| Operator<br>Variable                                                                                                                                                                                                                                                                                                                                   | Join-D<br>(leave blank) and (leave blank)                                                  |   |                                                                                                          |
| Operator<br>Variable                                                                                                                                                                                                                                                                                                                                   | Join-E<br>(leave blank) ("Wrong!")                                                         |  |                                                                                                          |
| <p>Connect the Joins in the following manner:</p> <p>#1 Place Join-B inside Join-A (right side)</p> <p>#2 Place Join-C in Join-D (right side)</p> <p>#3 Place Join-A in Join-D (left side)</p> <p>#4 Place Join-D in Join-E (left side)</p> <p><b>DUPLICATE</b> entire join string for Script #2<br/>Change text at the end (E) to <b>Correct!</b></p> |                                                                                            |  |                                                                                                          |

|                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Looks (2)                                                                                                                                                                                                                      | Say ( ) for ( ) seconds<br>one for each Join string                                                                                                                                                                                                                                                                                                   | This entire join string will fit inside the<br>“Say ( )” section of this Looks block. | For a recorded message that reads:<br><br>14 + 5 = 20 Wrong!! |
|  <p>The Join strings will look like this when done. There are <b>FIVE join</b> blocks and <b>THREE variables</b>.</p>                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |                                                               |
| <p>One Join block string will go <b>INSIDE</b> the first “then” loop - between the two variable blocks.</p> <p>One Join Stack will go <b>OUTSIDE</b> the second “else” loop at the bottom -- inside the first “else” loop.</p> |                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |                                                               |
| Control<br><br>Looks<br>Operators(5)<br>Variables(3)<br><br>Control<br>Control<br><br>Looks<br>Operators(5)<br>Variables(3)                                                                                                    | <p>if, then ...</p> <p><b>INSIDE “if, then” statement</b></p> <p>say (E-D-A-B-C string (Correct!))<br/>for (1) seconds</p> <p>... else</p> <p>if, then ... else</p> <p><b>OUTSIDE the else statement</b></p> <p>say (E-D-A-B-C string (Wrong!))<br/>for (1) seconds</p> <p>NOTE: It is outside the second else statement... but inside the first.</p> |    |                                                               |